

To understand a concept, a person must understand some collection of information and a framework by which to interpret the information. With a given amount of time and effort to invest, I think of efficiently learning as a 2 step process:

(1) Consider broadly the relative importance of different pieces of information and ordering (does learning certain information *first* make it easier to learn other information) and consider the relative importance of different elements of a framework. As an example, in calculus, understanding the spatial interpretation of a derivative is more important in a high school applied physics setting than understanding hyperbolic trigonometry.

(2) Take steps towards learning the content, weighted by the relative importance. In this case, spend more time earlier understanding derivative interpretation and less time later on hyperbolic trig. The former is a central focal point of understanding the subject as a whole and the latter is a tangential side-topic of little practical relevance.

I think people, or at least students in high school, allocate far too much time and effort in the second step and far too little in the first. There are diminishing returns to the first, but I think many people move on before they've even gotten close to an optimal allocation, and then proceed in a deeply inefficient way.

I think this general concept is true of many things. It's like a poorly planned military operation where troops are gunned down while relentlessly pushing towards a fortified bunker, or electing to do a survey for a prize rather than briefly analyzing the probability of winning and estimated completion time before making the decision to do it.

In many aspects of life a little bit of foresight is worth a lot of effort, and I think learning is one where this should be more emphasized. I feel that even a 10 minute discussion of this at the beginning of certain high school classes could make a world of difference to students that are working hard but still struggling to succeed.